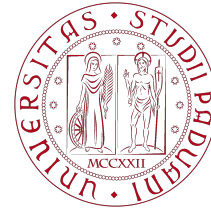


Final Report

Physics of Complex Networks: Structure and Dynamics



UNIVERSITÀ
DEGLI STUDI
DI PADOVA

Areas of physics by complexity



Newton's
Mechanics

Electro-
Magnetism

Special
Relativity

Quantum Mechanics
General Relativity

Quantum
Field Theory

Complexity
Science

Projects Report: #15, #47

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Last update: August 5, 2026

Contents

1	Task #15: Cascade Failures - SOC model	1
1.1	Introduction:	1
1.2	Another section...	2
2	Task title...	5
2.1	A section...	5
2.2	Another section...	5
3	Bibliography	6

List of Figures

1.1	Avalanche size and lifetime distributions for the SOC sandpile model. .	2
1.2	Avalanche size and lifetime distributions for the SOC sandpile model. .	3
1.3	Avalanche size and lifetime distributions for the SOC sandpile model. .	4
1.4	Avalanche size and lifetime distributions for the SOC sandpile model. .	4

1 | Task #15: Cascade Failures - SOC model

Structure as¹:

- A short (max 1 page) explanation of the task, including references. Include mathematical concepts.
- Max 2 pages for the whole task (including figures)
- It is possible to use appendices for supplementary material, at the end of the report. Max 5 pages per task

A total of 3 pages + 5 supplementary pages per task

1.1 | Introduction:

Self-Organized Criticality (SOC) is an approach adopted to explain the emergence of power-law distributions in natural phenomena. In physics, criticality is associated to systems near phase transitions, that live between two distinct regimes, and at this critical point, the emergence of power-laws occurs.

In SOC, systems are believed to organize themselves naturally into critical states, following internal feedback mechanisms and without external intervention.

In this report, we analyze the SOC behavior associated to the Sandpile Model.

In the **Sandpile Model**, we consider networks with an associated load variable, that accounts for any sort of stress that can be observed in real applications (such as power grids). A given node in the network stays functional, as long as its load doesn't go above a given threshold: its *capacity*. When the capacity of a node is overcome, we have a failure in the node, and its load gets redistributed according to some chosen rules.

We will use different types of networks, and observe the emergence of power-law distribution for the size A of the avalanches and their lifetime T . Particularly, we will consider different degree distributions:

- Gaussian: $k \sim \mathcal{N}(\mu = 20, \sigma = 8)$
- Uniform: $k \sim \text{Uniform}(k_{\min} = 4, k_{\max} = 36)$

¹Remove this part from the report

- Power Law (Scale Free): $k \sim k^{-\gamma}$ with $\gamma \in \{2.01, 2.2, 2.4, 2.6, 2.8, 3.0, 5.0\}$

Every network had 10^6 nodes, and the reported results are the averages of 30 iterations.

1.2 | Another section...

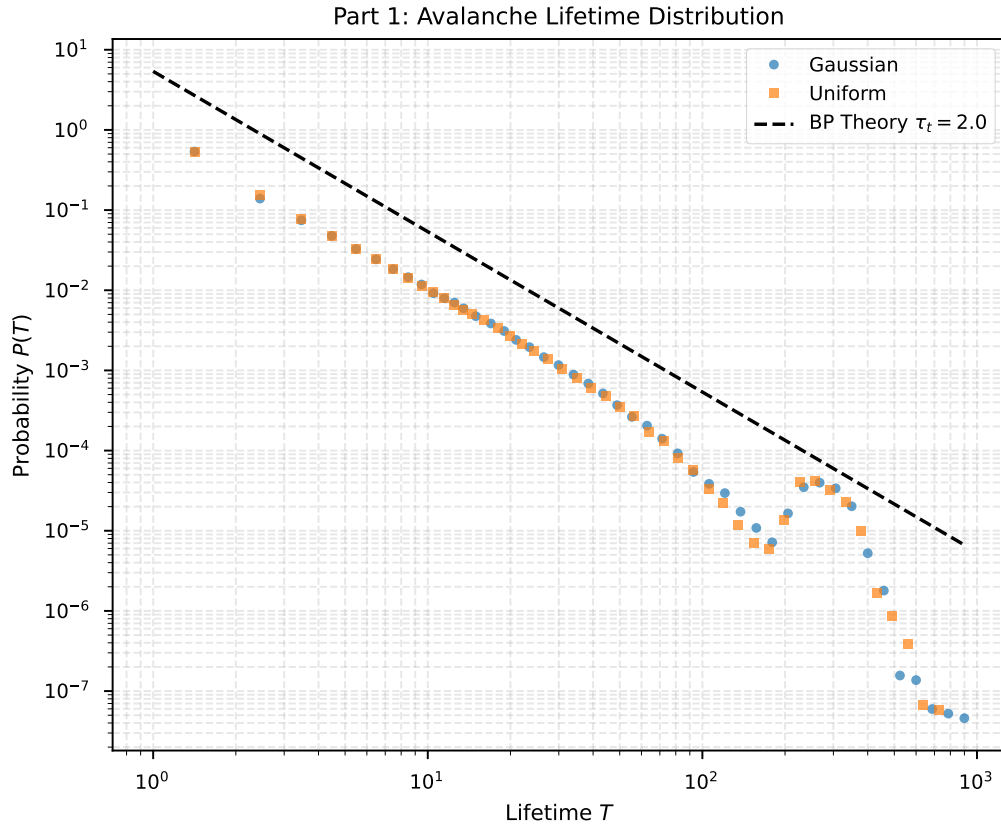


Figure 1.1: Avalanche size and lifetime distributions for the SOC sandpile model.

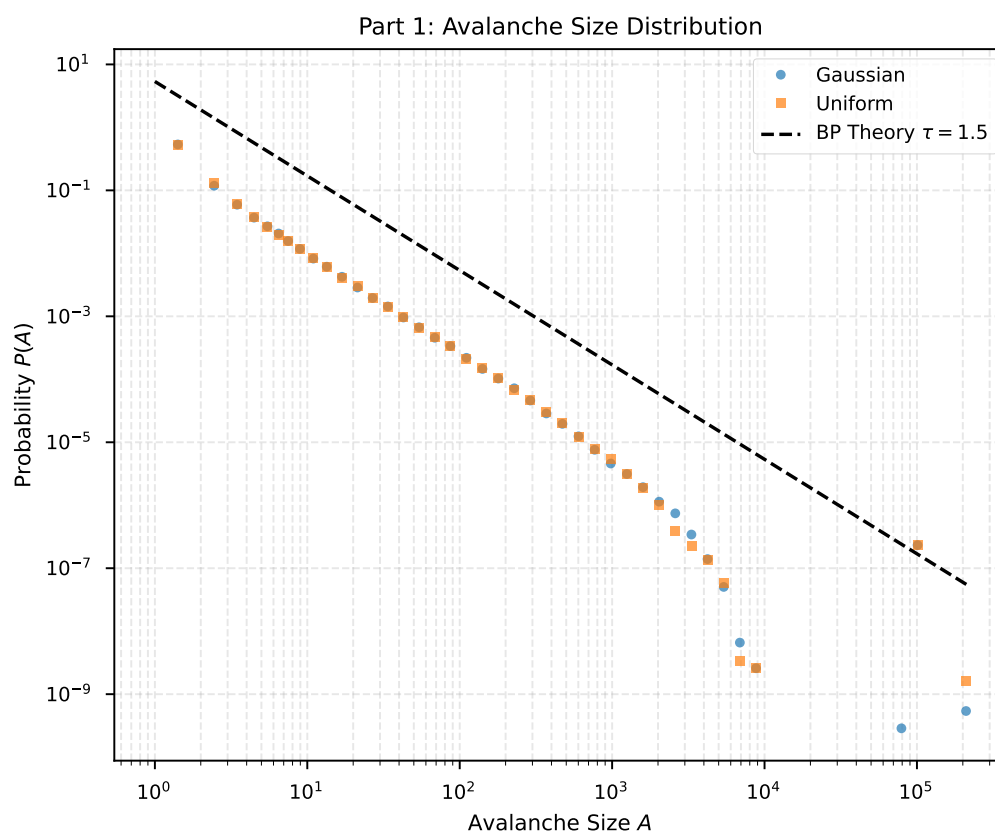


Figure 1.2: Avalanche size and lifetime distributions for the SOC sandpile model.

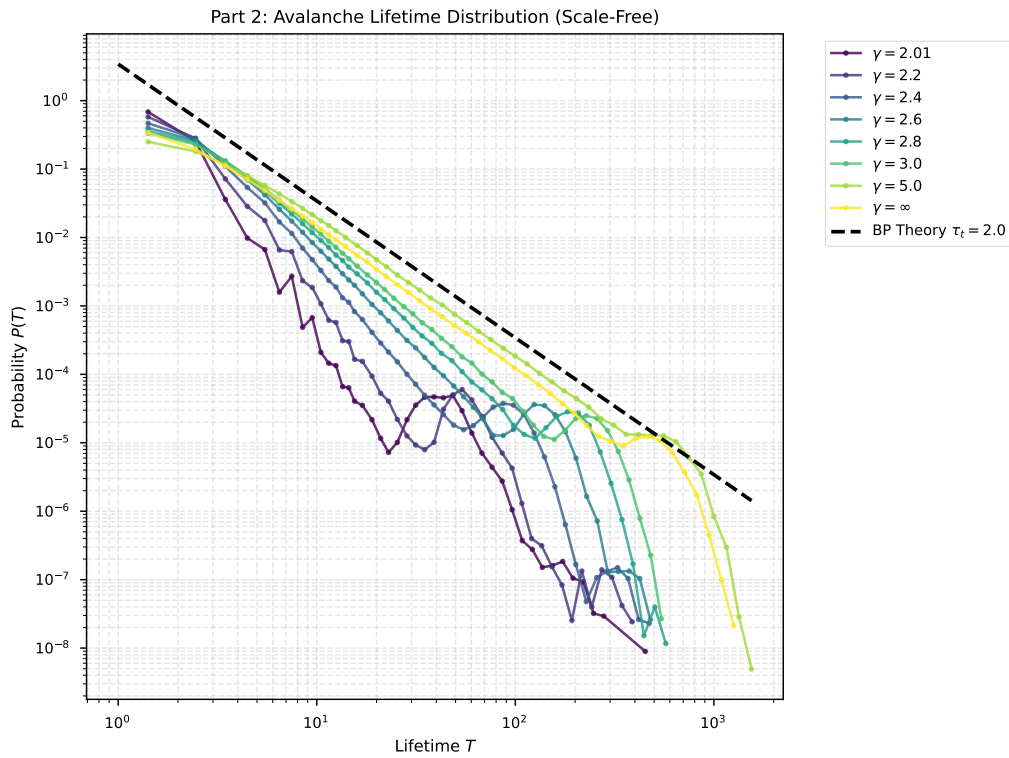


Figure 1.3: Avalanche size and lifetime distributions for the SOC sandpile model.

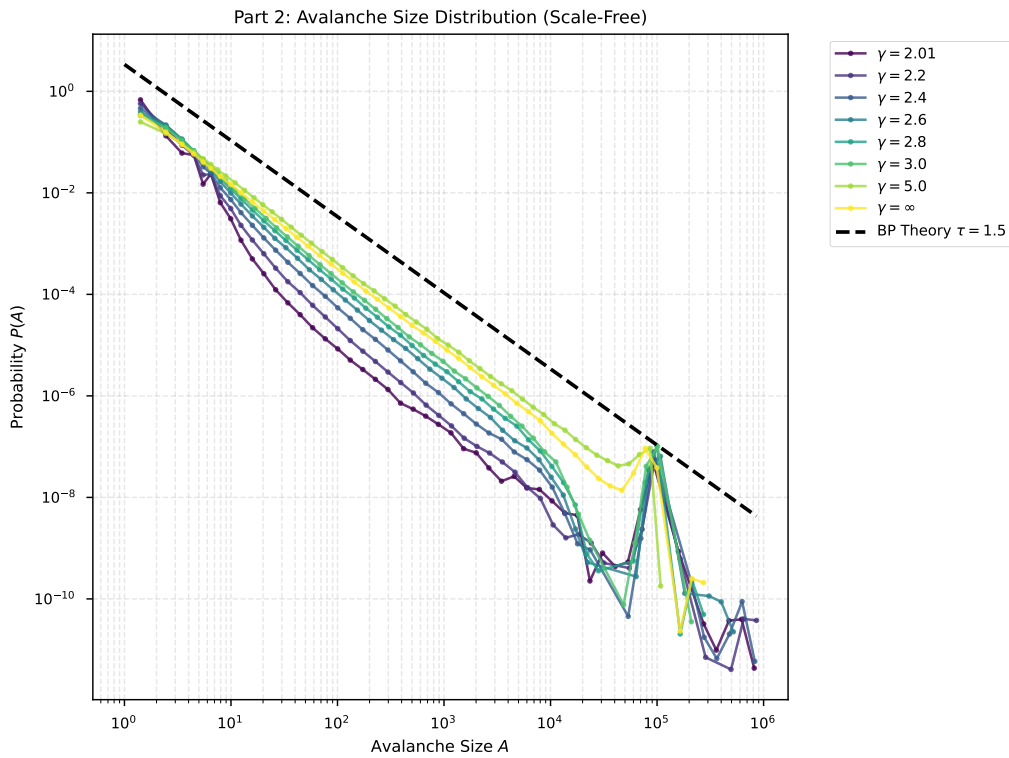


Figure 1.4: Avalanche size and lifetime distributions for the SOC sandpile model.

2 | Task title...

Task leader(s): *Author name(s)...*

Structure as¹:

- *A short (max 1 page) explanation of the task, including references. Include mathematical concepts.*
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A total of 3 pages + 5 supplementary pages per task

2.1 | A section...

Reference examples: book [2], article [3], website [1]

2.2 | Another section...

¹Remove this part from the report

3 | Bibliography

- [1] Protecting critical infrastructure from a cyber pandemic — weforum.org. <https://www.weforum.org/agenda/2021/10/protecting-critical-infrastructure-from-cyber-pandemic/>. [Accessed 29-Mar-2023].
- [2] Alain Barrat, Marc Barthélemy, and Alessandro Vespignani. *Dynamical processes on complex networks*. Cambridge university press, 2008.
- [3] Manlio De Domenico, Albert Solé-Ribalta, Emanuele Cozzo, Mikko Kivelä, Yamir Moreno, Mason A Porter, Sergio Gómez, and Alex Arenas. Mathematical formulation of multilayer networks. *Physical Review X*, 3(4):041022, 2013.